PVC Suspension Reactor Ver 1 · R-101 (one of four 140 m³ batch suspension-PVC reactors)

Plot plan, elevations and piping isometrics (R-101 area)

PVC-V1-PL-LAY-001

Drawing set of the R-101 area: plot plan at scale with 18 spacing rules (18 met), bay plan, elevation with the NPSH and gravity-discharge heights, 6 piping isometrics with bills of materials, the coordinate table that layout.json holds, the 12-station operator route and 45 checks (43 ✓ / 2 ✗).

Unit	PVC Suspension Reactor Ver 1 · R-101 (one of four 140 m³ batch suspension-PVC reactors)
Prepared for	Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University
Document no.	PVC-V1-PL-LAY-001
Title	Plot plan, elevations and piping isometrics (R-101 area)
Status	Rev 0 · Issued for teaching
Originator	Apinan Soottitantawat

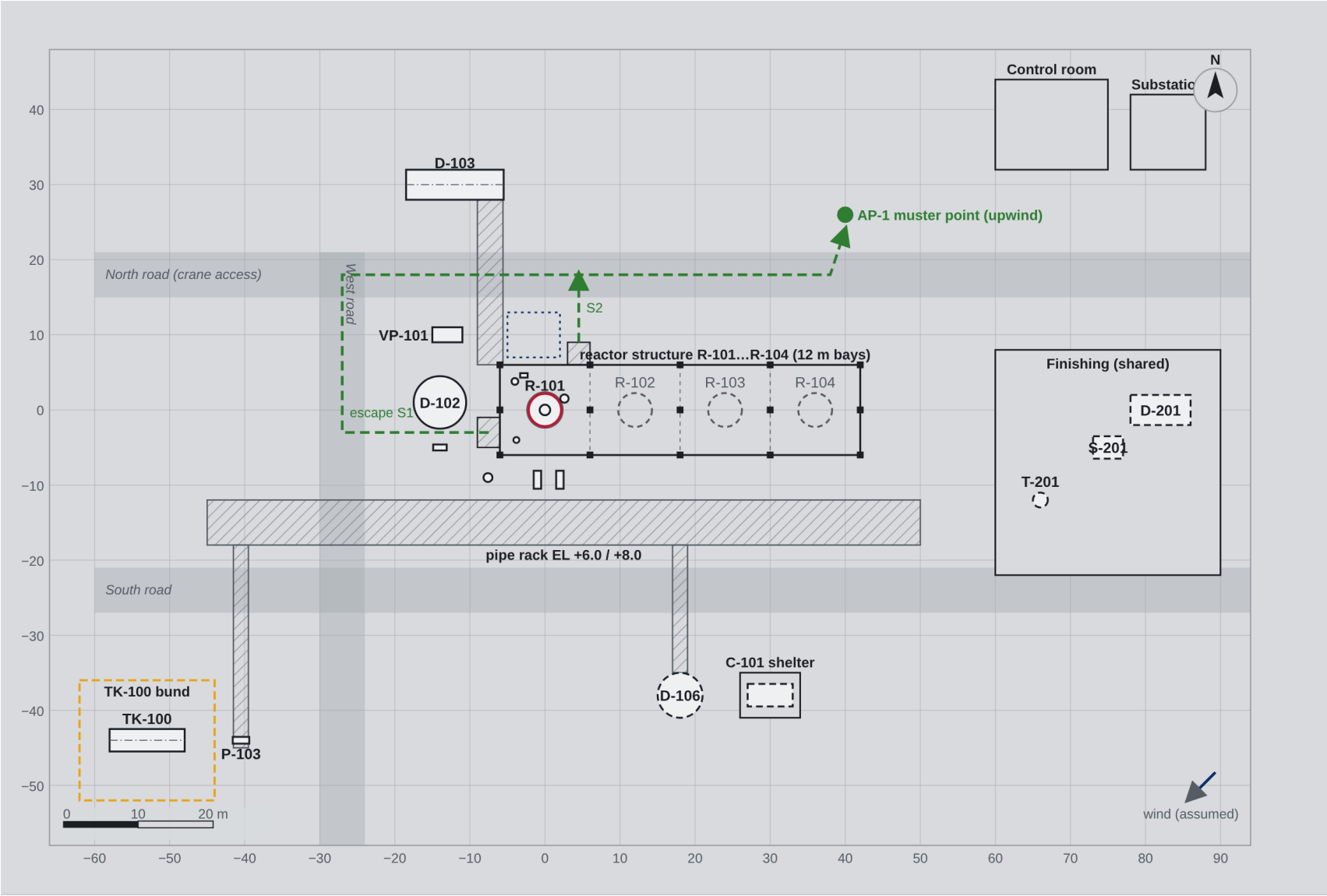
Rev	Date	Description	Originator	Checked	Approved
0	2026-10-27	Issued for teaching	Apinan Soottitantawat		

The numbers in this document come from the PVC Suspension Reactor Ver 1 simulator model (R-101) and the documents of the same OTS, or are labelled "assumed" or "typical". It was generated for teaching: it is not a design deliverable and must not be used for construction or purchase. Checked and Approved are left blank on purpose.

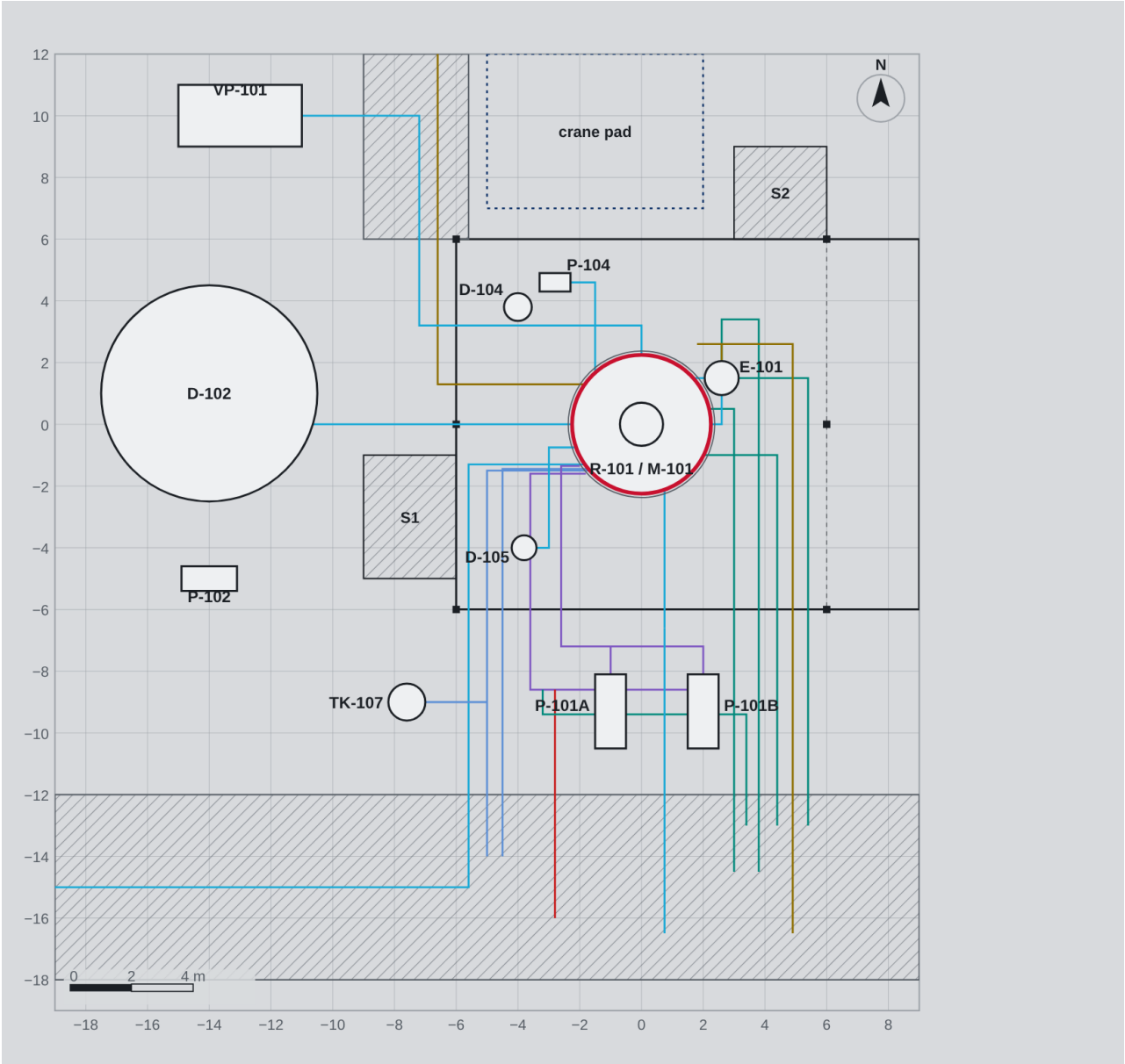
1 Scope

Plot plan, R-101 bay plan, elevation, six piping isometrics with bills of materials, the coordinate table, the outside-operator route and the checks of the R-101 area of the PVC Suspension Reactor Ver 1 OTS. Positions follow typical spacing rules (assumed); sizes come from the equipment list PVC-V1-ME-EQL-001 and PFD-100, line numbers and sizes from PVC-V1-PE-HMB-001. The same data is held in layout.json, which the Visual plant 3D model reads.

2 Plot plan



Plot plan of the R-101 area (grid 10 m from the R-101 centreline; prevailing wind assumed from the north-east).



R-101 bay, all levels (grid 2 m).

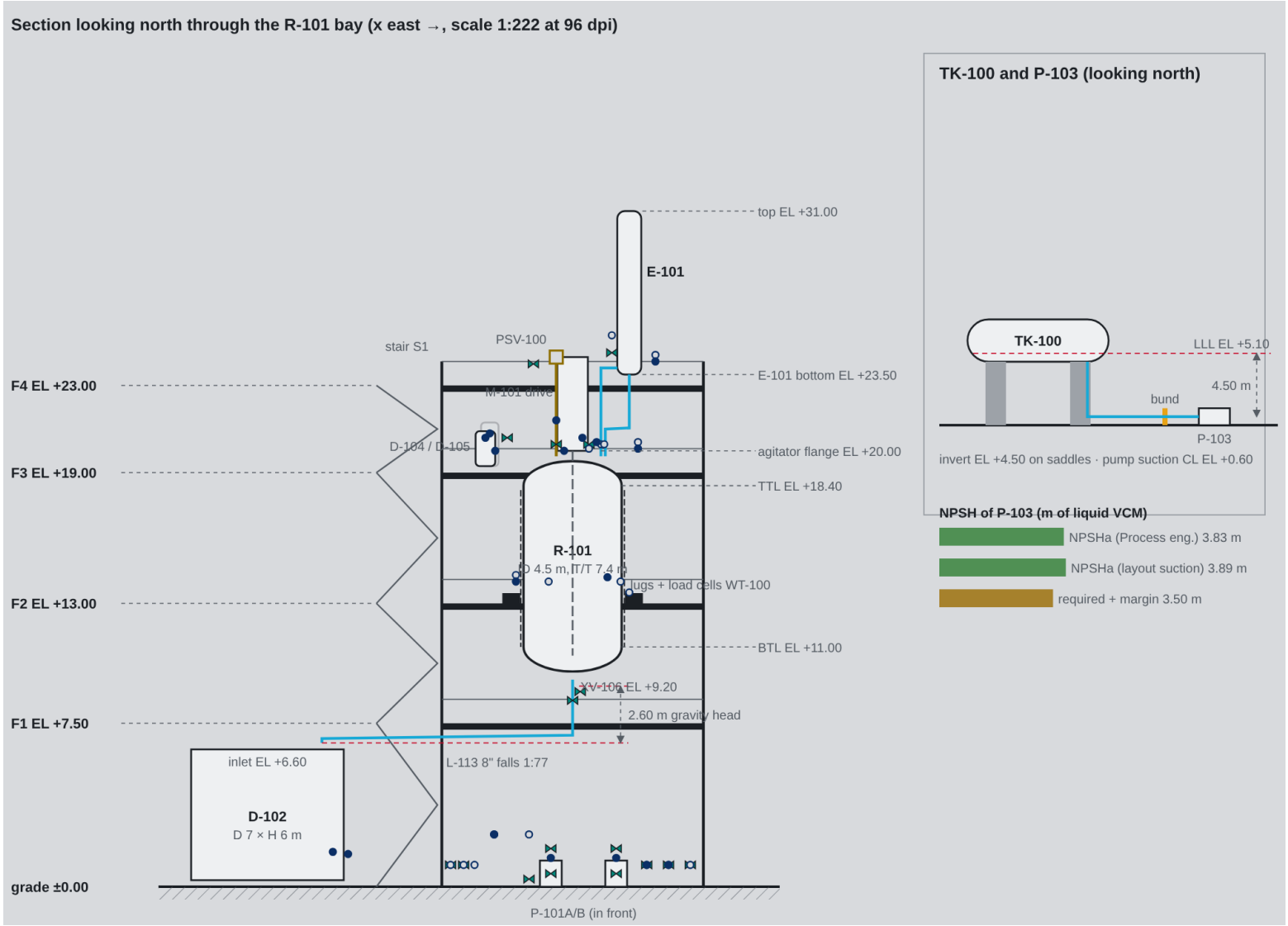
2.1 Spacing and siting rules (typical, assumed)

Rule	Between	Result	Min.	OK	Basis
S1	Pressurised liquefied flammable storage (TK-100) to the process unit (reactor structure)	55.6 m	30 m	✓	typical 30 m (CCPS / GAPS-type spacing tables for LPG-type pressure storage to process units), assumed
S2	TK-100 to the control room (occupied building)	131.2 m	60 m	✓	typical 60 m for pressure storage to occupied buildings, assumed
S3	TK-100 to the substation / MCC (ignition source)	146.4 m	30 m	✓	typical 30 m, assumed
S4	TK-100 to the nearest road edge (vehicles are ignition sources)	15.5 m	15 m	✓	typical 15 m, assumed
S5	TK-100 to fired equipment (D-201 hot-air heater)	132.3 m	30 m	✓	typical 30 m (fired heaters to flammable storage), assumed
S6	P-103 outside the TK-100 bund and ≥ 3 m from the tank shell	outside the bund, 6.4 m to the shell	—	✓	pumps for flammable liquids outside impounding areas (NFPA 30-type practice), assumed
S7	Control room to the reactor structure	31.6 m	30 m	✓	typical 30 m for a blast-resistant occupied building (≥ 60 m if not blast-rated; building siting study API RP 752-type), assumed
S8	Substation / MCC (ignition source) to the reactor structure	44.4 m	15 m	✓	typical 15 m (outside the hazardous area), assumed
S9	Fired equipment (D-201 hot-air heater) to VCM inventories (reactor structure, D-106, D-103, TK-100)	nearest RS-1 36.0 m	30 m	✓	typical 30 m, upwind or crosswind, assumed
S10	Compressor C-101 to other process equipment (its own suction vessel D-106 excepted)	nearest RS-1 30.5 m	9 m	✓	typical 9 m (30 ft) for compressors handling flammable gas, assumed
S11	Gasholder D-106 to the reactor structure	29.0 m	15 m	✓	typical 15 m, assumed
S12	Gasholder D-106 to the control room	78.6 m	30 m	✓	typical 30 m, assumed
S13	Relief receiver D-103 to the control room	65.5 m	30 m	✓	typical 30 m, assumed
S14	Reactor to reactor, shell to shell (jacket included)	7.3 m between jackets	4.5 m	✓	≥ one diameter for access and maintenance, assumed
S15	No pump below a vessel (P-101A/B, P-102, P-103, P-104, VP-101)	none of 6 pumps below a vessel	—	✓	pumps leak and catch fire; keep them out from under vessels and structures holding flammables (typical practice)
S16	Crane access for agitator and E-101 removal: hardstanding within 16 m of the agitator axis, hook height for the shaft	radius 10.1 m, hook height needed 36.8 m	—	✓	agitator flange +20.0 + shaft 9.5 m + drive 4.3 m + rigging 3 m ≤ 45 m hook height of the assumed mobile crane
S17	Fire-fighting access: every equipment item within 30 m of a road edge	farthest D-105 16.6 m	—	✓	typical hose reach 30 m, assumed
S18	Two escape routes from every R-101 platform, in different directions, travel ≤ 25 m to a stair	4 of 4 platforms	—	✓	typical practice (two means of escape from elevated platforms), assumed

2.2 Coordinates and levels

Tag	Name	Shape	x m	y m	z m	Top m	Footprint m	Shared
TK-100	VCM Storage	hcyl	−53.00	−44.00	4.50	7.50	10.0 × 3.0	
P-103	VCM Charge Pump	pump	−40.50	−44.00	0.00	1.20	2.2 × 0.9	
TK-107	PVA Pot	vcyl	−7.60	−9.00	0.30	2.10	Ø 1.20	
D-104	Initiator Pot (chilled)	vcyl	−4.00	3.80	19.30	20.90	Ø 0.90	
P-104	Initiator Dosing Pump	pump	−2.80	4.60	19.00	19.70	1.0 × 0.6	
D-105	Short-stop Pot (N2)	vcyl	−3.80	−4.00	19.30	21.30	Ø 0.80	
R-101	PVC Reactor 140 m3	vcyl	0.00	0.00	9.88	19.52	Ø 4.50	
R-101 J	R-101 jacket	vcyl	0.00	0.00	11.00	18.20	Ø 4.75	
M-101	R-101 agitator	motor	0.00	0.00	20.00	24.30	Ø 1.40	
E-101	Reflux Condenser	exchanger	2.60	1.50	23.50	31.00	Ø 1.10	
VP-101	Vacuum Pump	pump	−13.00	10.00	0.00	1.80	4.0 × 2.0	
D-103	Emergency Vent Tank	hcyl	−12.00	30.00	0.60	4.60	13.0 × 4.0	
D-106	VCM Gasholder 1.25 bar a	holder	18.00	−38.00	0.30	5.30	Ø 6.00	yes
C-101	VCM Recovery Compressor	box	30.00	−38.00	0.00	2.50	6.0 × 3.0	yes
P-101A	Jacket Circulation Pump A	pump	−1.00	−9.30	0.00	1.20	1.0 × 2.4	
P-101B	Jacket Circulation Pump B	pump	2.00	−9.30	0.00	1.20	1.0 × 2.4	
D-102	Blowdown / Slurry Tank	vcyl	−14.00	1.00	0.30	6.30	Ø 7.00	
P-102	Slurry Pump	pump	−14.00	−5.00	0.00	1.00	1.8 × 0.8	
T-201	Slurry Stripping Column	vcyl	66.00	−12.00	2.00	16.00	Ø 2.00	yes
S-201	Dewatering Centrifuge	box	75.00	−5.00	0.00	3.00	4.0 × 3.0	yes
D-201	PVC Dryer	box	82.00	0.00	0.00	5.00	8.0 × 4.0	yes
R-102	PVC Reactor 140 m3 (identical)	vcyl	12.00	0.00	9.88	19.52	Ø 4.50	
R-103	PVC Reactor 140 m3 (identical)	vcyl	24.00	0.00	9.88	19.52	Ø 4.50	
R-104	PVC Reactor 140 m3 (identical)	vcyl	36.00	0.00	9.88	19.52	Ø 4.50	

3 Elevations and levels



Section through the R-101 bay looking north, and TK-100 / P-103 with the NPSH of P-103.

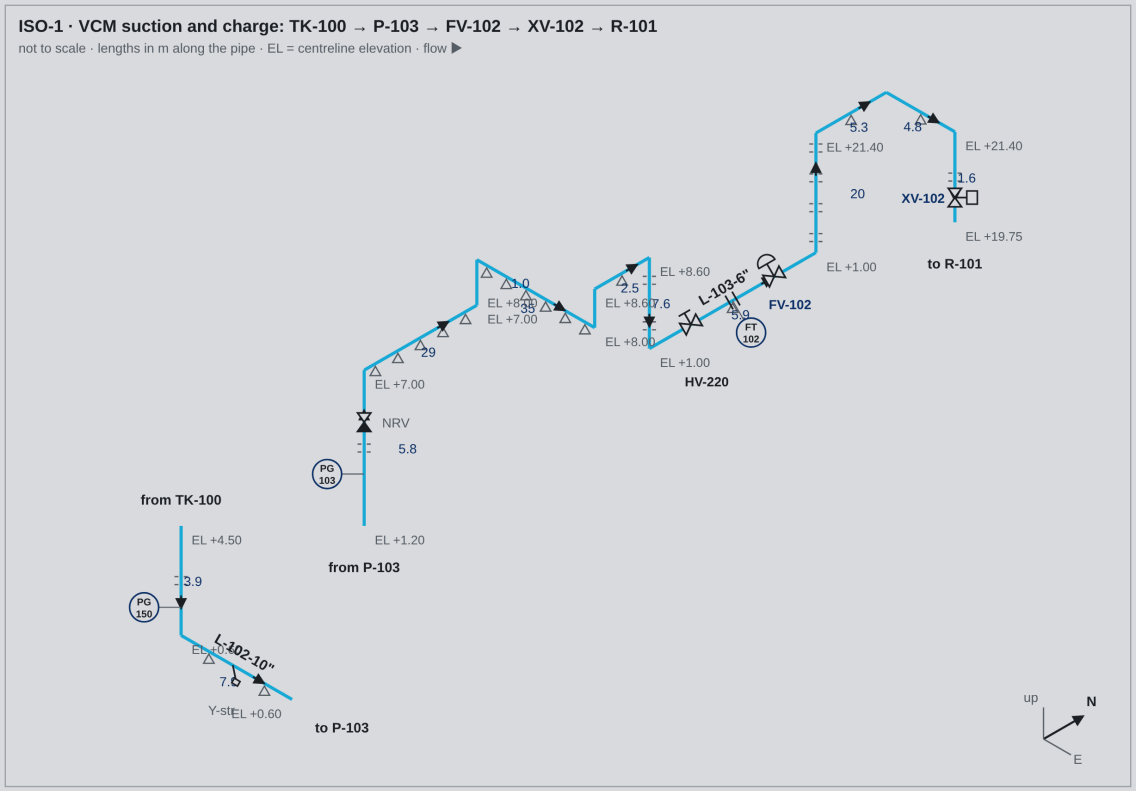
3.1 Where gauges and hand valves are read or operated from

Tag	Item	Elevation	Standing level	Height	Range	OK
HV-201	hand valve (NO)	EL +0.60	grade	0.60 m	0.3–1.9 m	✓
HV-202	hand valve (NO)	EL +1.75	grade	1.75 m	0.3–1.9 m	✓
HV-203	hand valve (NO)	EL +0.60	grade	0.60 m	0.3–1.9 m	✓
HV-204	hand valve (NO)	EL +1.75	grade	1.75 m	0.3–1.9 m	✓
HV-210	hand valve (NO)	EL +1.00	grade	1.00 m	0.3–1.9 m	✓
HV-211	hand valve (NO)	EL +1.00	grade	1.00 m	0.3–1.9 m	✓
HV-212	hand valve (NO)	EL +1.00	grade	1.00 m	0.3–1.9 m	✓
HV-220	hand valve (NO)	EL +1.00	grade	1.00 m	0.3–1.9 m	✓
HV-221	hand valve (NO)	EL +1.00	grade	1.00 m	0.3–1.9 m	✓
HV-230	hand valve (NO)	EL +20.30	F3 EL +19.00	1.30 m	0.3–1.9 m	✓
HV-231	hand valve (NO)	EL +24.50	F4 EL +23.00	1.50 m	0.3–1.9 m	✓
HV-240	hand valve (NO)	EL +8.55	F1 EL +7.50	1.05 m	0.3–1.9 m	✓
HV-250	hand valve (LO)	EL +20.30	F3 EL +19.00	1.30 m	0.3–1.9 m	✓
HV-251	hand valve (LO)	EL +23.99	F4 EL +23.00	0.99 m	0.3–1.9 m	✓
HV-252	hand valve (LO)	EL +20.60	F3 EL +19.00	1.60 m	0.3–1.9 m	✓
HV-260	hand valve (LC)	EL +8.95	F1 EL +7.50	1.45 m	0.3–1.9 m	✓
HV-261	hand valve (LC)	EL +0.35	grade	0.35 m	0.3–1.9 m	✓
HV-107	hand valve (NO)	EL +1.60	grade	1.60 m	0.3–1.9 m	✓
HV-141	hand valve (NO)	EL +20.30	F3 EL +19.00	1.30 m	0.3–1.9 m	✓
FT-101	transmitter (indicator)	EL +1.00	grade	1.00 m	0.5–2.5 m	✓
FT-102	transmitter (indicator)	EL +1.00	grade	1.00 m	0.5–2.5 m	✓
FT-103	transmitter (indicator)	EL +1.00	grade	1.00 m	0.5–2.5 m	✓
PT-100	transmitter (indicator)	EL +20.30	F3 EL +19.00	1.30 m	0.5–2.5 m	✓
PT-101	transmitter (indicator)	EL +20.30	F3 EL +19.00	1.30 m	0.5–2.5 m	✓
LT-100	transmitter (indicator)	EL +20.10	F3 EL +19.00	1.10 m	0.5–2.5 m	✓
TT-100	transmitter (indicator)	EL +14.00	F2 EL +13.00	1.00 m	0.5–2.5 m	✓
TT-101	transmitter (indicator)	EL +14.00	F2 EL +13.00	1.00 m	0.5–2.5 m	✓
TT-111	transmitter (indicator)	EL +14.30	F2 EL +13.00	1.30 m	0.5–2.5 m	✓
TT-110	transmitter (indicator)	EL +2.40	grade	2.40 m	0.5–2.5 m	✓
FT-110	transmitter (indicator)	EL +2.40	grade	2.40 m	0.5–2.5 m	✓
FT-120	transmitter (indicator)	EL +1.00	grade	1.00 m	0.5–2.5 m	✓
TT-121	transmitter (indicator)	EL +20.40	F3 EL +19.00	1.40 m	0.5–2.5 m	✓
FT-130	transmitter (indicator)	EL +1.00	grade	1.00 m	0.5–2.5 m	✓
TT-131	transmitter (indicator)	EL +24.40	F4 EL +23.00	1.40 m	0.5–2.5 m	✓
TT-132	transmitter (indicator)	EL +25.30	F4 EL +23.00	2.30 m	0.5–2.5 m	✓
TT-120	transmitter (indicator)	EL +1.00	grade	1.00 m	0.5–2.5 m	✓

Tag	Item	Elevation	Standing level	Height	Range	OK
WT-100	transmitter (indicator)	EL +13.50	F2 EL +13.00	0.50 m	0.5–2.5 m	✓
PG-100	local gauge	EL +20.40	F3 EL +19.00	1.40 m	0.5–2.5 m	✓
TG-100	local gauge	EL +14.20	F2 EL +13.00	1.20 m	0.5–2.5 m	✓
SG-100	local gauge	EL +20.00	F3 EL +19.00	1.00 m	0.5–2.5 m	✓
PG-101A	local gauge	EL +1.32	grade	1.32 m	0.5–2.5 m	✓
PG-101B	local gauge	EL +1.32	grade	1.32 m	0.5–2.5 m	✓
TG-110	local gauge	EL +2.40	grade	2.40 m	0.5–2.5 m	✓
TG-111	local gauge	EL +14.00	F2 EL +13.00	1.00 m	0.5–2.5 m	✓
TG-120	local gauge	EL +1.00	grade	1.00 m	0.5–2.5 m	✓
PG-120	local gauge	EL +1.00	grade	1.00 m	0.5–2.5 m	✓
TG-121	local gauge	EL +20.10	F3 EL +19.00	1.10 m	0.5–2.5 m	✓
TG-131	local gauge	EL +24.10	F4 EL +23.00	1.10 m	0.5–2.5 m	✓
PG-102	local gauge	EL +1.50	grade	1.50 m	0.5–2.5 m	✓
PG-103	local gauge	EL +1.60	grade	1.60 m	0.5–2.5 m	✓
PG-150	local gauge	EL +1.60	grade	1.60 m	0.5–2.5 m	✓
LG-150	local gauge	EL +2.00	grade	2.00 m	0.5–2.5 m	✓
PG-104	local gauge	EL +20.60	F3 EL +19.00	1.60 m	0.5–2.5 m	✓
TG-104	local gauge	EL +20.00	F3 EL +19.00	1.00 m	0.5–2.5 m	✓
PG-105	local gauge	EL +20.80	F3 EL +19.00	1.80 m	0.5–2.5 m	✓
PG-141	local gauge	EL +20.60	F3 EL +19.00	1.60 m	0.5–2.5 m	✓
PG-250	local gauge	EL +21.40	F3 EL +19.00	2.40 m	0.5–2.5 m	✓
PG-106	local gauge	EL +1.50	grade	1.50 m	0.5–2.5 m	✓
PG-180	local gauge	EL +1.50	grade	1.50 m	0.5–2.5 m	✓
LG-180	local gauge	EL +1.60	grade	1.60 m	0.5–2.5 m	✓
PG-109	local gauge	EL +1.60	grade	1.60 m	0.5–2.5 m	✓

4 Piping isometrics

ISO-1 VCM suction and charge: TK-100 → P-103 → FV-102 → XV-102 → R-101

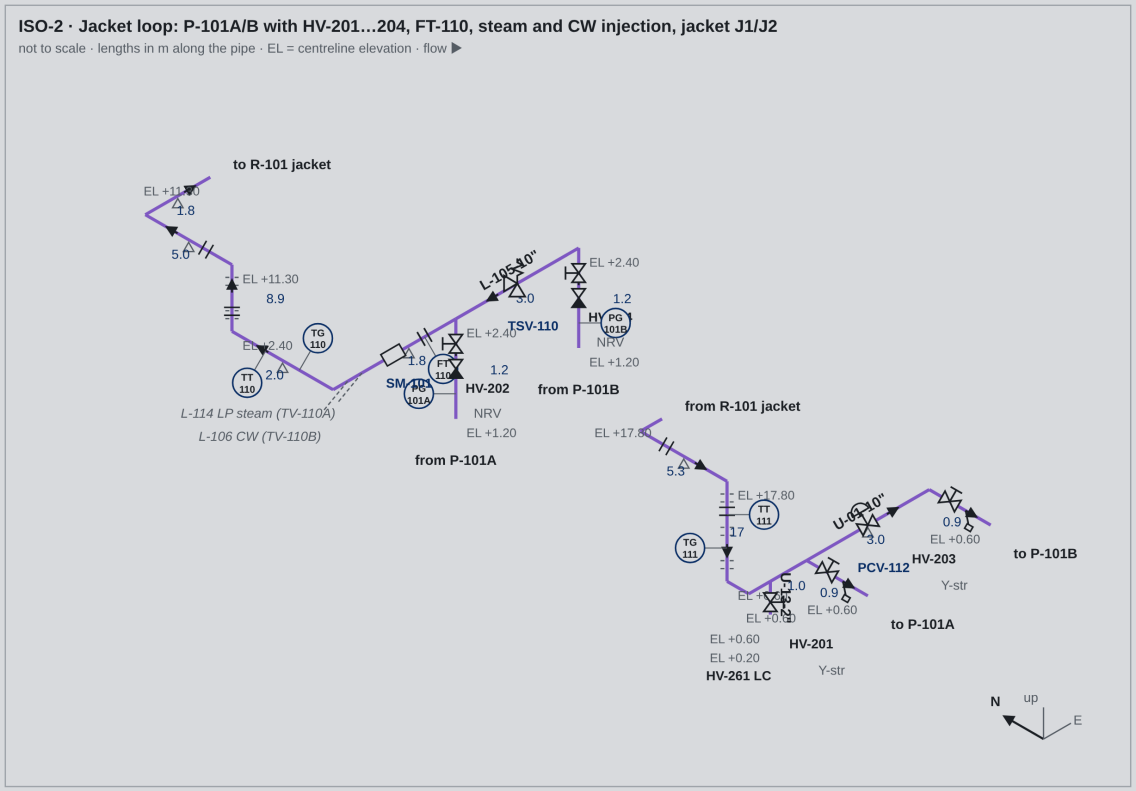


ISO-1 (not to scale; lengths in m).

Bill of materials	Quantity
Pipe 6"	120.0 m
Pipe 10"	11.8 m
Elbows	12
Tees	0
Hand valves	1
Control valves	1
On-off valves	1
Check valves	1
Relief devices	0
Flow elements	1

Bill of materials	Quantity
Instrument connections	2
Pipe flanges	16
Supports	26

ISO-2 Jacket loop: P-101A/B with HV-201…204, FT-110, steam and CW injection, jacket J1/J2

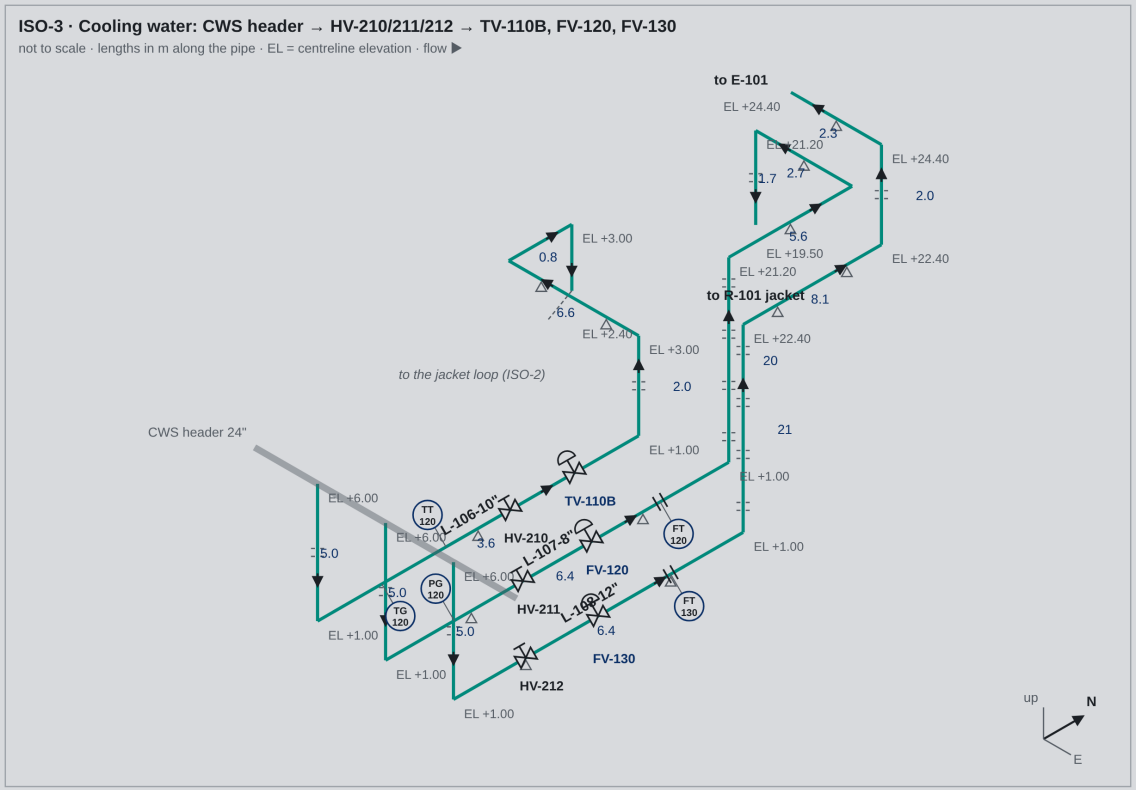


ISO-2 (not to scale; lengths in m).

Bill of materials	Quantity
Pipe 2"	0.4 m
Pipe 10"	55.8 m
Elbows	10
Tees	5
Hand valves	5
Control valves	1
On-off valves	0
Check valves	2
Relief devices	1
Flow elements	1

Bill of materials	Quantity
Instrument connections	6
Pipe flanges	32
Supports	11

ISO-3 Cooling water: CWS header → HV-210/211/212 → TV-110B, FV-120, FV-130

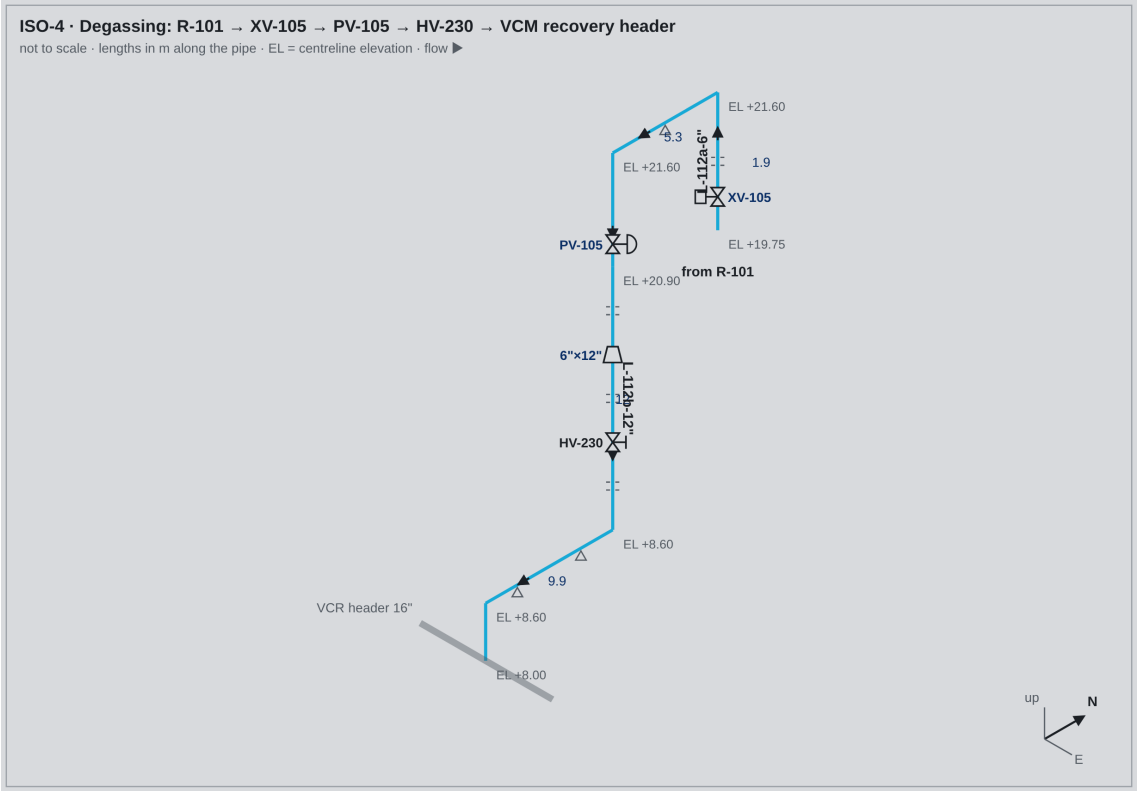


ISO-3 (not to scale; lengths in m).

Bill of materials	Quantity
Pipe 8"	41.6 m
Pipe 10"	18.6 m
Pipe 12"	45.1 m
Elbows	15
Tees	1
Hand valves	3
Control valves	3
On-off valves	0
Check valves	0
Relief devices	0

Bill of materials	Quantity
Flow elements	2
Instrument connections	3
Pipe flanges	18
Supports	26

ISO-4 Degassing: R-101 → XV-105 → PV-105 → HV-230 → VCM recovery header

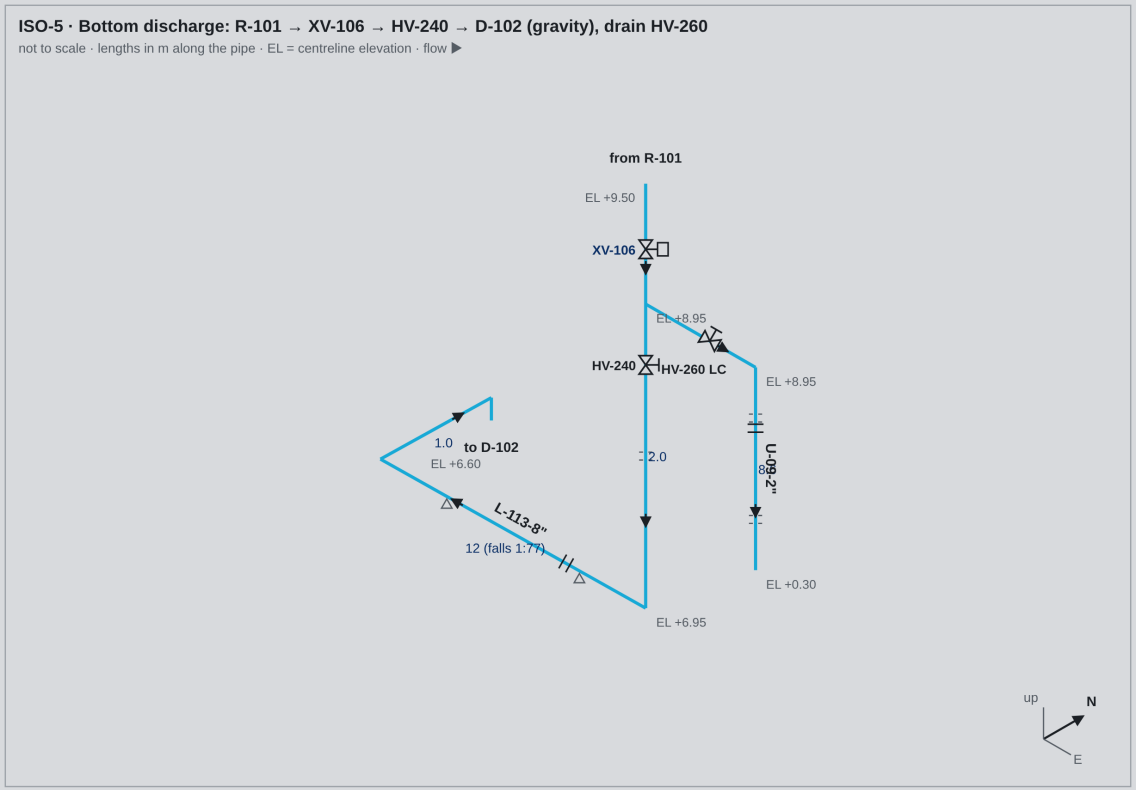


ISO-4 (not to scale; lengths in m).

Bill of materials	Quantity
Pipe 6"	7.9 m
Pipe 12"	22.8 m
Elbows	4
Tees	0
Hand valves	1
Control valves	1
On-off valves	1
Check valves	0
Relief devices	0
Flow elements	0

Bill of materials	Quantity
Instrument connections	0
Pipe flanges	7
Supports	7

ISO-5 Bottom discharge: R-101 → XV-106 → HV-240 → D-102 (gravity), drain HV-260

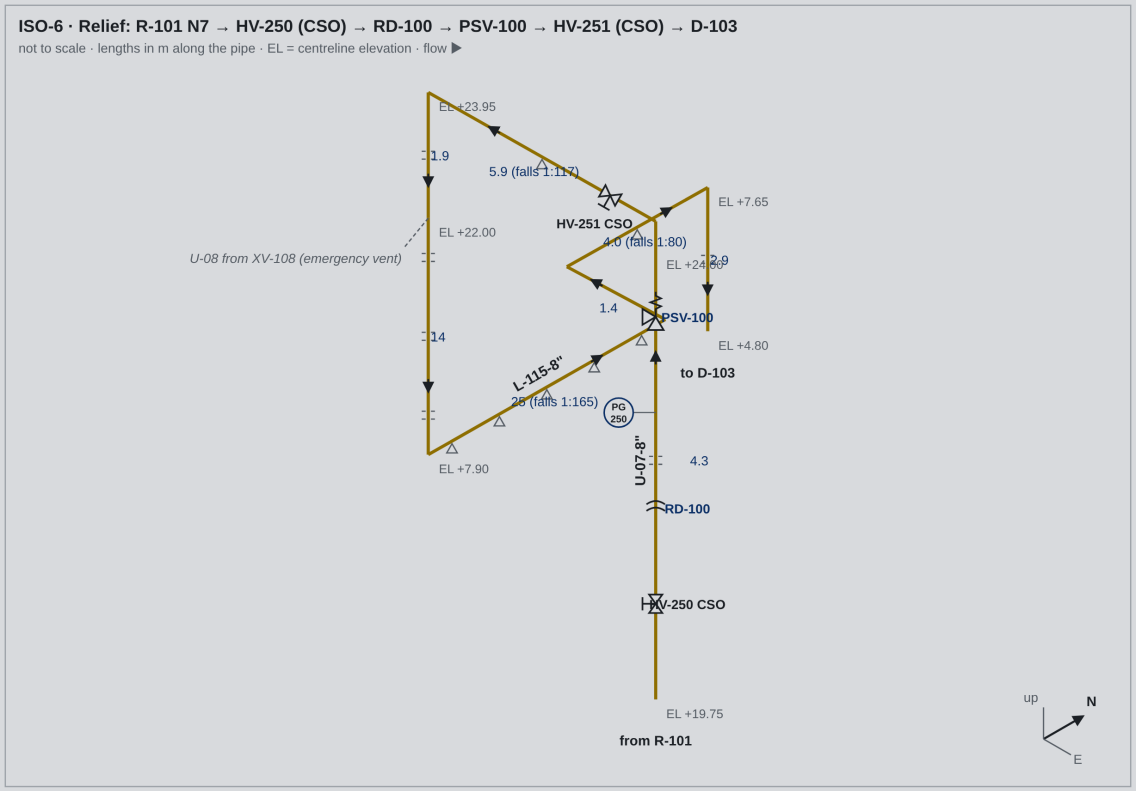


ISO-5 (not to scale; lengths in m).

Bill of materials	Quantity
Pipe 2"	9.2 m
Pipe 8"	15.2 m
Elbows	4
Tees	1
Hand valves	2
Control valves	0
On-off valves	1
Check valves	0
Relief devices	0
Flow elements	0

Bill of materials	Quantity
Instrument connections	0
Pipe flanges	8
Supports	5

ISO-6 Relief: R-101 N7 → HV-250 (CSO) → RD-100 → PSV-100 → HV-251 (CSO) → D-103

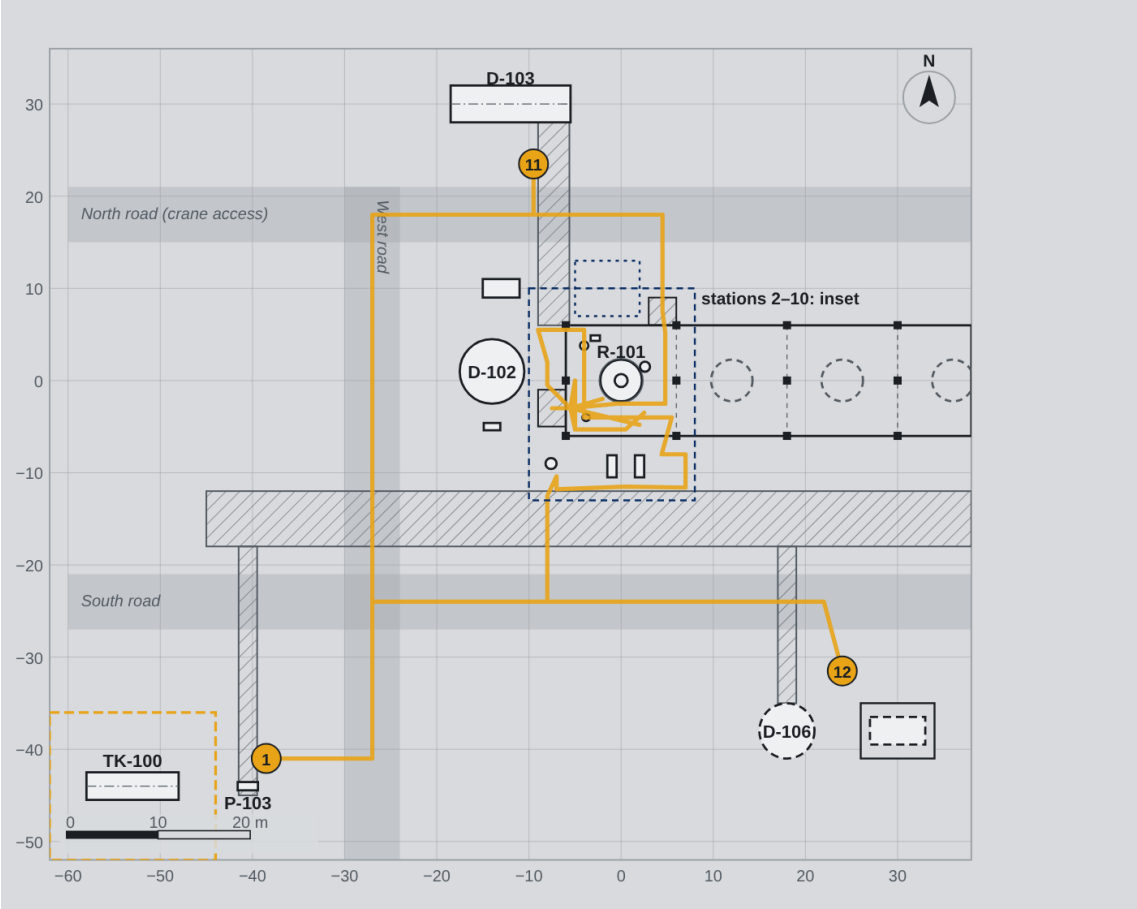


ISO-6 (not to scale; lengths in m).

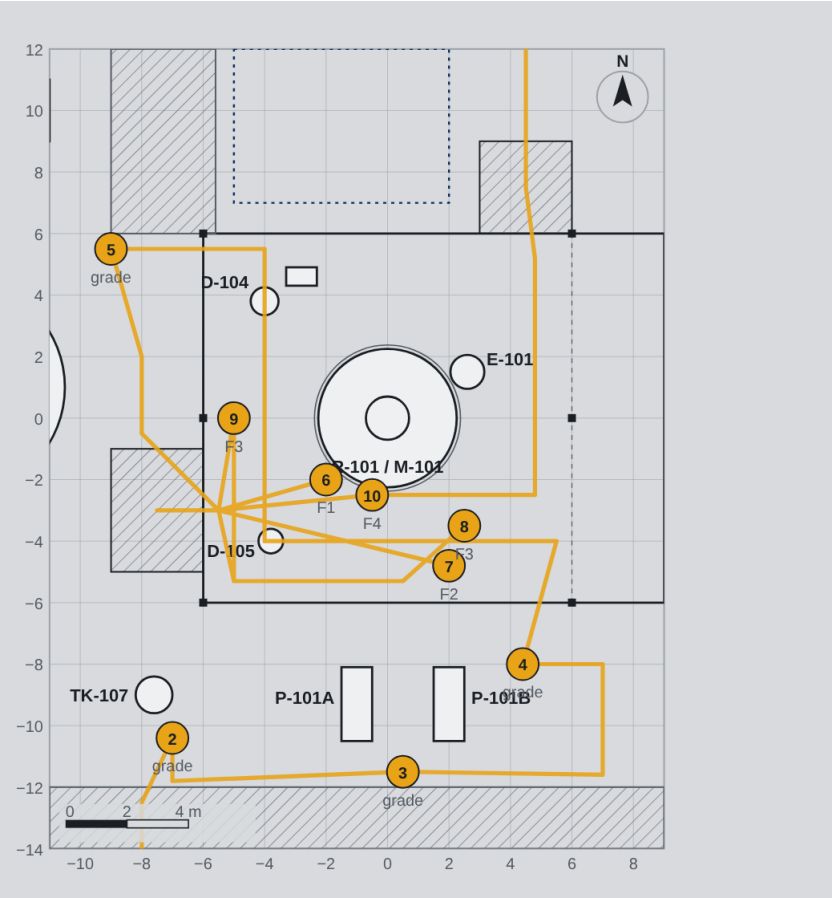
Bill of materials	Quantity
Pipe 8"	59.1 m
Elbows	5
Tees	1
Hand valves	2
Control valves	0
On-off valves	0
Check valves	0
Relief devices	2
Flow elements	0
Instrument connections	1

Bill of materials	Quantity
Pipe flanges	10
Supports	12

5 Outside-operator route



Route on the plot plan.



Stations inside the R-101 bay.

Stn	Name	Items
1	VCM storage TK-100 and charge pump P-103	TK-100, LG-150, PG-150, P-103, PG-103
2	Charge station at grade: FV-101, FV-102, FV-103 and the PVA pot	HV-220, FT-102, FV-102, HV-221, FT-101, FV-101, FT-103, FV-103, TK-107, HV-107
3	Jacket-loop pumps P-101A/B	P-101A, P-101B, HV-201, HV-202, HV-203, HV-204, PG-101A, PG-101B, FT-110, TSV-110, PCV-112, HV-261, TV-110A, TT-110, TG-110
4	Cooling-water station: TV-110B, FV-120, FV-130	HV-210, HV-211, HV-212, TV-110B, FV-120, FV-130, FT-120, FT-130, TT-120, TG-120, PG-120
5	D-102 blowdown tank and VP-101 (grade, west and north-west)	D-102, PG-102, P-102, VP-101, PG-109
6	F1 EL +7.5 · bottom valve XV-106 and the line to D-102	XV-106, HV-240, HV-260
7	F2 EL +13.0 · jacket, thermowells and load cells	TT-100, TT-101, TG-100, WT-100, TT-111, TG-111

Stn	Name	Items
8	F3 EL +19.0 · top head: charge, degassing, vent and relief nozzles	M-101, PT-100, PT-101, PG-100, LT-100, SG-100, XV-101, XV-102, XV-105, PV-105, HV-230, XV-108, XV-109, PCV-141, PG-141, HV-141, HV-250, RD-100, PG-250, TT-121, TG-121
9	F3 · initiator pot D-104 and short-stop pot D-105	D-104, P-104, PG-104, TG-104, XV-103, D-105, PG-105, HV-252, XV-104
10	F4 EL +23.0 · reflux condenser E-101 and PSV-100	E-101, TT-131, TG-131, TT-132, HV-135, HV-231, PSV-100, HV-251
11	Relief receiver D-103 (north of the road)	D-103, PG-180, LG-180
12	VCM recovery D-106 and C-101 (shared, south of the road)	D-106, PG-106, C-101

6 Checks

43 ✓ and 2 ✗ of 45 checks, computed from layout.json and the other documents. The ✗ are raised as engineering queries.

#	Check	Value	Criterion	OK	Source
1	Every EM_EQL item is placed on the plot plan	20 of 20	all	✓	EM_EQL (Mechanical)
2	Footprints and heights agree with the PFD-100 real sizes	5 of 5 (TK-100, R-101, D-106, D-102, T-201)	equal	✓	PF_SPEC.real
3	Every line of the line list is routed in layout.json	16 of 16	all	✓	EP_HMB.lines
4	Pipe sizes in layout.json = line-list sizes	15 of 15 (L-115 has no size on the line list: 8" assumed)	equal	✓	EP_HMB.lines
5	Every line number on an isometric exists in EP_HMB.lines	10 of 10 (L-102, L-103, L-105, L-106, L-107, L-108, L-112a, L-112b, L-113, L-115)	all	✓	EP_HMB.lines
6	Every spool drawn on an isometric carries a line number	4 spools without a number: U-01 (Jacket return J2 → P-101A/B suction header); U-13 (Jacket loop low-point drain); U-09 (R-101 bottom drain → closed drain); U-07 (PSV-100 inlet spool R-101 N7 → HV-250 → RD-100 → PSV-100)	0	✗	line list vs isometrics → EQ-LA-01
7	Every hand valve of HANDVALVES.json is placed (same key and normal position)	17 of 17	all	✓	HANDVALVES.json
8	Every hand valve sits on the pipe of its line	17 of 17	all	✓	layout.json pipes
9	Every hand valve is in a route station	17 of 17	all	✓	route
10	Every hand valve can be operated from its standing level (0.3–1.9 m above it)	19 of 19	all	✓	part 2.3
11	Every field instrument of the instrument list is placed (transmitters, valves, regulators, switches)	40 of 40	all	✓	EI_INL
12	Every local gauge and transmitter is placed and in a route station	42 of 42 (24 gauges, 18 transmitters)	all	✓	route
13	Every local gauge and transmitter is readable (0.5–2.5 m above its standing level)	42 of 42	all	✓	part 2.3
14	Every "reads" key is a live model value (sim.meas)	27 of 27	all	✓	sim.meas
15	Spacing rule S1: Pressurised liquefied flammable storage (TK-100) to the process unit (reactor structure)	55.6 m	≥ 30 m	✓	part 1.2 (typical, assumed)
16	Spacing rule S2: TK-100 to the control room (occupied building)	131.2 m	≥ 60 m	✓	part 1.2 (typical, assumed)
17	Spacing rule S3: TK-100 to the substation / MCC (ignition source)	146.4 m	≥ 30 m	✓	part 1.2 (typical, assumed)
18	Spacing rule S4: TK-100 to the nearest road edge (vehicles are ignition sources)	15.5 m	≥ 15 m	✓	part 1.2 (typical, assumed)
19	Spacing rule S5: TK-100 to fired equipment (D-201 hot-air heater)	132.3 m	≥ 30 m	✓	part 1.2 (typical, assumed)
20	Spacing rule S6: P-103 outside the TK-100 bund and ≥ 3 m from the tank shell	outside the bund, 6.4 m to the shell	see rule	✓	part 1.2 (typical, assumed)
21	Spacing rule S7: Control room to the reactor structure	31.6 m	≥ 30 m	✓	part 1.2 (typical, assumed)
22	Spacing rule S8: Substation / MCC (ignition source) to the reactor structure	44.4 m	≥ 15 m	✓	part 1.2 (typical, assumed)
23	Spacing rule S9: Fired equipment (D-201 hot-air heater) to VCM inventories (reactor structure, D-106, D-103, TK-100)	nearest RS-1 36.0 m	≥ 30 m	✓	part 1.2 (typical, assumed)
24	Spacing rule S10: Compressor C-101 to other process equipment (its own suction vessel D-106 excepted)	nearest RS-1 30.5 m	≥ 9 m	✓	part 1.2 (typical, assumed)
25	Spacing rule S11: Gasholder D-106 to the reactor structure	29.0 m	≥ 15 m	✓	part 1.2 (typical, assumed)
26	Spacing rule S12: Gasholder D-106 to the control room	78.6 m	≥ 30 m	✓	part 1.2 (typical, assumed)

#	Check	Value	Criterion	OK	Source
27	Spacing rule S13: Relief receiver D-103 to the control room	65.5 m	≥ 30 m	✓	part 1.2 (typical, assumed)
28	Spacing rule S14: Reactor to reactor, shell to shell (jacket included)	7.3 m between jackets	≥ 4.5 m	✓	part 1.2 (typical, assumed)
29	Spacing rule S15: No pump below a vessel (P-101A/B, P-102, P-103, P-104, VP-101)	none of 6 pumps below a vessel	see rule	✓	part 1.2 (typical, assumed)
30	Spacing rule S16: Crane access for agitator and E-101 removal: hardstanding within 16 m of the agitator axis, hook height for the shaft	radius 10.1 m, hook height needed 36.8 m	see rule	✓	part 1.2 (typical, assumed)
31	Spacing rule S17: Fire-fighting access: every equipment item within 30 m of a road edge	farthest D-105 16.6 m	see rule	✓	part 1.2 (typical, assumed)
32	Platform F1 EL +7.50 has two exits	2 stairs (S1 west, S2 north), 90° apart, longest travel 15.0 m	2 exits, ≥ 90°, ≤ 25 m	✓	layout.json platforms
33	Platform F2 EL +13.00 has two exits	2 stairs (S1 west, S2 north), 90° apart, longest travel 15.0 m	2 exits, ≥ 90°, ≤ 25 m	✓	layout.json platforms
34	Platform F3 EL +19.00 has two exits	2 stairs (S1 west, S2 north), 90° apart, longest travel 15.0 m	2 exits, ≥ 90°, ≤ 25 m	✓	layout.json platforms
35	Platform F4 EL +23.00 has two exits	2 stairs (S1 west, S2 north), 90° apart, longest travel 15.0 m	2 exits, ≥ 90°, ≤ 25 m	✓	layout.json platforms
36	layout.json loads and validates against the contract	0 errors	0 errors	✓	STEP16_BRIEF contract; node tools/test.js
37	layout.json is the generator output (no hand edits)	identical to laBuild(HANDVALVES.json)	identical	✓	tools/mklayout.js
38	Route: 12 stations, eyes 1.6 m above the floor stood on	12 stations	10–14	✓	layout.json route
39	P-103 static head from the layout = the Process engineering assumption	4.50 m vs EP_P103.z 4.50 m	equal	✓	EP_P103
40	P-103 NPSH available with the layout suction line ≥ required + margin	3.89 m (L-102 11.8 m; Process engineering 3.83 m with 30 m)	≥ 3.50 m	✓	epP103()
41	P-103 discharge in the layout = the hydraulics assumption (static rise, length)	rise 18.55 m vs 10 m; L-103 120 m vs 80 m	equal ±10 %	✗	EP_P103 → EQ-LA-02
42	FV-102 keeps enough pressure drop at the highest reactor pressure with the layout discharge	2.49 bar (Process engineering 3.41 bar)	≥ 0.5 bar	✓	epP103()
43	Bottom discharge L-113 falls all the way to D-102	min slope 1:77, XV-106 2.60 m above the D-102 inlet	≥ 1:100, no rise	✓	part 2
44	PSV-100 discharge L-115 is free-draining to D-103 (no pocket)	falls from EL +24.00 to EL +4.80	no rise	✓	API 521-type practice
45	E-101 sits above R-101 so the reflux runs back by gravity	E-101 bottom EL +23.50 vs nozzles EL +19.75	higher	✓	part 2